

Crowdsensed Drive Test Application

Stakeholder Interview — RF / Drive Test Engineer

Stakeholder: Drive Test / RF Engineer

To capture expert knowledge of drive test methodology, tools, and pain points for replication via smartphone.

University of Buea · Dr. Valery Nkemeni · April 11, 2026

Informed Consent & Confidentiality Notice

This questionnaire is part of an academic research project at the University of Buea (CEF440). Your participation is entirely voluntary. All responses will be used solely for academic purposes and will be kept strictly confidential. No personally identifiable information will be published. By completing this questionnaire, you consent to its use in this study. Thank you for your time.

Respondent Information

Name (optional):

Organisation:

Role / Specialisation:

Years in RF/Drive Testing:

Date:

Section A: Drive Test Methodology

Q1.

What professional drive test tools or software do you currently use?

☐ TEMS Investigation

☐ NEMO Outdoor

☐ G-NetTrack

☐ Accuver XCAL

☐ Rohde & Schwarz ROMES

☐ NetMonster (Android)

☐ Custom/In-house tool

☐ Other (specify)

Q2.

Walk us through your standard drive test procedure from start to finish. What steps do you follow?

Q3.

What network parameters do you always collect during a drive test? List them in order of importance.

Q4.

What is the typical sampling interval you use when collecting measurements (e.g., every 1 second, every 10 metres)?

Q5.

How do you handle GPS accuracy issues during data collection? What threshold of GPS accuracy do you accept?

Section B: Limitations of Traditional Drive Testing

Q6.

What are the biggest challenges or frustrations with traditional drive testing? (Select all that apply)

- | | |
|---|---|
| ☐ High equipment cost | ☐ Time-consuming route planning |
| ☐ Difficult terrain / inaccessible areas | ☐ Requires dedicated personnel |
| ☐ Cannot scale to cover all areas | ☐ Data is a snapshot in time, not continuous |
| ☐ Vehicle access limitations | ☐ Other (specify) |

Q7.

Estimate the average cost (time + equipment + personnel) of a single full drive test campaign in your area.

Q8.

Are there areas in your network zone that have never been physically drive-tested? Why?

Section C: Smartphone as a Measurement Device

Q9.

In your professional opinion, how accurately can a modern Android smartphone replicate measurements from dedicated drive test equipment?

Very inaccurately ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 *Very accurately*

Q10.

Which parameters do you believe a smartphone CAN reliably measure? (Select all that apply)

- | | |
|--|---|
| ☐ RSRP | ☐ RSRQ |
| ☐ RSSI | ☐ SINR |
| ☐ Cell ID | ☐ Network Type (2G/3G/4G/5G) |
| ☐ GPS coordinates | ☐ Throughput (speed test) |
| ☐ Timing Advance | |

Q11.

Which parameters do you believe a smartphone CANNOT reliably measure compared to professional tools?

Q12.

What minimum number of crowdsourced data points per km² would you consider statistically meaningful for coverage analysis?

Section D: Data Quality & Validation

Q13.

How would you validate the quality of crowdsourced data before using it in network planning decisions?

Q14.

What data cleaning or filtering steps should be applied to crowdsourced measurements? (e.g., removing indoor measurements, filtering by speed)

Q15.

Should the app distinguish between measurements taken while stationary vs. moving? How should this be handled?

☐ Yes — only collect while moving above a speed threshold

☐ No — collect regardless of motion

☐ Yes — tag measurements with motion state but collect both

☐ Not sure

Q16.

What metadata should accompany each measurement record to make it useful for professional analysis?

Section E: Feature Recommendations

Q17.

If you were designing this application, what are the three most important features you would include?

Q18.

Would you personally be willing to use or recommend such an app to your team as a supplementary tool?

- ☐ Yes — as a full replacement for drive tests in some scenarios
- ☐ Yes — as a supplementary/complementary tool only
- ☐ Maybe — depends on demonstrated accuracy
- ☐ No — I would not trust smartphone-based data

Q19.

Any additional technical recommendations or warnings for the development team?

Thank you for your expert input. Your insights are invaluable to this project.